The Quantum Random Oracle Toolbox

Hans Heum

NTNU - Norwegian University of Science and Technology, Trondheim, Norway.
hans.heum@ntnu.no

Abstract. In which we reveal the tricks of the trade.

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1 Introduction

There are three main challenges for proofs in the quantum random oracle model, relating to the application of standard techniques in the classical random oracle model:

1. Recording: or, how to extract QRO queries.
2. Reprogramming: or, how to alter specific outputs of the QRO.
3. Rewinding: or, how to derandomize quantum oracle algorithms.

2 Preliminaries

Basic notation. We take the natural numbers $\mathbb{N}$ to include 0. For $n \in \mathbb{N}$, we define the set $[n] = \{0, \ldots, n-1\}$. We will employ the convention that $N = 2^n$; thus, $NM = 2^{n+m}$.

Functions are defined over bitstrings unless otherwise noted, and the set of all functions from the set of m-length bitsrings $\{0,1\}^m$ to the set of n-length $\{0,1\}^n$ bitstrings is denoted $\mathcal{F}_{m \to n}$.

We use $\mathbb{1}_{n \times n}$ to denote the n-by-n identity matrix, dropping the subscript whenever the dimension is clear from the context.

3 A Brief Quantum Primer

Quantum theory arose as a branch of physics, and so much of the standard notational conventions are hand-wavy, and often inconsistently implemented across the literature. For the sake of clarity and self-containedness, we specify all our conventions below, providing a brief summary of the building blocks that make up quantum computing theory along the way; for more details, see the celebrated introductory text by Nielsen and Chuang [NC10].

Dirac notation. In general, quantum states are complex-valued vectors in a Hilbert space (a vector space defined over functions), which may or may not be finite-dimensional.[1] Quantum states are represented using the Dirac notation, in which a quantum state carrying the *label* ψ is written $|\psi\rangle$, its conjugate transpose is written $\langle\psi| \coloneqq |\psi\rangle^\dagger \coloneqq (|\psi\rangle^*)^T$, and its inner product with itself is written $\langle\psi|\psi\rangle \coloneqq ||\psi\rangle|^2$. Quantum states are unit-norm, and so we have that $\langle\psi|\psi\rangle = 1$. For two not necessarily equal quantum states $|\psi\rangle$ and $|\varphi\rangle$, their inner product $\langle\psi|\varphi\rangle$ is a real number between zero and one, where "zero" means the states are orthogonal, and "one" means the states are identical (modulo an unobservable global phase factor $e^{i\theta}$).

This calculus is fine, but one may wonder, what *is* a quantum state—in the sense, what kind of mathematical object is it? The answer can be found in the statement that they live in N-dimensional *Hilbert space:* Hilbert spaces are vector spaces defined over *functions* in place of vectors, where the inner product between two functions f and g on the interval $[a,b]$ is defined to be $\int_{-\infty}^{\infty} f(x)^* g(x) dx$. This inner product defines a norm, and a notion of orthogonality, from which a vector space may be constructed. Quantum states, then, are the functions that arise as solutions to a differential equation describing the dynamics of the system, i.e. the Schrödinger equation in the non-relativistic, non-spin case; the Pauli equation in the case with spin (such as electrons), and the Dirac equation in the relativistic case (excluding gravity). They take some input variable corresponding to an observable, such as the position x, and returns the *amplitude* corresponding to that position; squaring this then gives the probability that observing the particle will yield x as a result.

If all of this sounds terribly complicated, then you understand why physicists grew so fond of Dirac's notation, which neatly swept all of these horrible details under the rug of conceptual clarity!

Thankfully, things become a bit easier in the discrete case: the Hilbert space is now finite, and so it can be expressed in terms of a finite set of basis "vectors" (i.e., functions), and the inner product is now a sum rather than an integral. Further, functions now have finite domains, and may thus be written as N-length vectors instead, where each element corresponds to an input to the function, enabling us to employ our toolbox from linear algebra to study them. This is the approach taken in the quantum information sciences.

[1] For example, consider the state $|p(x)\rangle$ describing the position $x \in [0,1]$ of particle p; this will live in an infinite-dimensional Hilbert space characterized by the basis vectors $\{|p(x)\rangle\}_{x \in [0,1]}$.

Qubit basis states. We are interested in finite-dimensional quantum states, implemented on a collection of n qubits. A *qubit* is a quantum state existing in the two-dimensional Hilbert space characterized by the basis states $|0\rangle$ and $|1\rangle$. In physical implementations, these states may for example be mapped to the spin of an electron ("up"/"down"), or the two lowest energy states of an atom, or in general any two orthogonal quantum states, where two quantum states are orthogonal if and only if there exists a measurement procedure that perfectly distinguishes them.

It is worth emphasizing that the "0" and "1" here are *labels*, and do not say anything about the physical system in question other than that it contains (at least) two orthogonal quantum states, which are being identified with the two basis states $|0\rangle$ and $|1\rangle$.

Since two qubit basis states $|b_1\rangle$ and $|b_2\rangle$, for $b_1, b_2 \in \{0,1\}$, the two states must be either identical (up to the aforementioned phase factor) or orthogonal, we have that

$$\langle b_1 | b_2 \rangle = \delta_{b_1 b_2} \,,$$

where δ_{ij} is the Kronecker delta, equal to 1 if $i = j$, and 0 otherwise.

Combined quantum states. In general, two quantum systems (i.e. states describing distinct physical systems) may be combined via the tensor product (also known as the *outer* product):

$$|\psi\rangle \otimes |\varphi\rangle \coloneqq |\psi\varphi\rangle \ .$$

If $|\psi\rangle$ describes an N-dimensional quantum system, and $|\varphi\rangle$ describes an M-dimensional quantum system, then the combined state $|\psi\varphi\rangle$ describing the combined system will be NM-dimensional. (Note how the dimensions are *multiplied* rather than added, as one might naïvely expect.)

Applying this to qubits, given a collection of n qubits $|b_i\rangle$, each initialized to one of the two basis states (so that $b_i \in \{0,1\}$), we may construct the combined state

$$|b_0 b_1 \dots b_{n-1}\rangle \coloneqq |b_0\rangle \otimes |b_1\rangle \otimes \dots \otimes |b_{n-1}\rangle \ .$$

Since each qubit is a two-dimensional quantum state, we see that the resulting combined state will have dimension $N \coloneqq 2^n$. The resulting Hilbert space will be characterized by the computational[2] basis states

$$\{|i\rangle\}_{\forall i \in [N]} = \{|b_0 b_1 \dots b_{n-1}\rangle\}_{\forall b_0, \dots, b_{n-1} \in \{0,1\}} \ .$$

This exponentiality in the dimension of quantum states as the number of qubits increases sits at the core of the quantum advantage over classical computers. As an illustrative example, the Hilbert space corresponding to a three-qubit quantum system is characterized by the eight computational basis states:

$$\begin{aligned}\{|i\rangle\}_{\forall i \in [2^3]} &= \{|000\rangle, |001\rangle, |010\rangle, |011\rangle, |100\rangle, |101\rangle, |110\rangle, |111\rangle\} \\ &= \left\{ \begin{bmatrix}1\\0\\0\\0\\0\\0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\\0\\0\\0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\\0\\0\\0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\\1\\0\\0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\\0\\1\\0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\\0\\0\\1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\\0\\0\\0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\\0\\0\\0\\0\\1\end{bmatrix} \right\} .\end{aligned}$$

As in the single-qubit case, we see that we have orthonormality over the computational basis states:

$$\langle x | y \rangle = \delta_{xy} \,.$$

Qubit registers. Given a collection of n qubits, we will often find it useful to divide it into *registers* of specified sizes, for example to separate distincts elements in an input. We will use subscripts to refer to specific registers, unless the register we are referring to is clear from context.

As an example, let's say we want the first n_1 qubits store the variable x—call this "Register 1"—and that the remaining n_2 qubits—"Register 2"—should store the variable y. Then the following are all equivalent ways of denoting the combined state:

$$|x \| y\rangle = |x\rangle_1 \otimes |y\rangle_2 \coloneqq |x\rangle_1 |y\rangle_2 \coloneqq |x, y\rangle \ ,$$

where $\|$ denotes string concatenation.

[2] As with any vector space, Hilbert spaces permit an infinitude of basis choices; the "computational" basis is simply the name of the choice wherein basis states are labelled by classical bits (or, in general, bitstrings).

Greek and Roman letters. We will use the letters x, y, z, ... to represent classical binary strings. Thus, a state $|x\rangle$ will always be understood to be a computational basis state. Whenever we are talking specifically about single-qubit computational basis states, we will use b to represent a single bit, and write $|b\rangle$.

General quantum states are traditionally labelled by letters taken from the middle of the greek alphabet, such as ψ, φ, ξ, etc. We will follow this tradition, or otherwise use a descriptive label relating to the specific context in which the state is being used.

Quantum *amplitudes*, meanwhile, are written using letters taken from the start of the greek alphabet (α, β, ...).

General qubit states. A general single-qubit state may be written in terms of the computational basis states as follows:

$$|\psi\rangle = \alpha_0 |0\rangle + \alpha_1 |1\rangle \ ,$$

where the amplitudes α_i are complex numbers satisfying $|\alpha_0|^2 + |\alpha_1|^2 = 1$, (where the absolute value of a complex number α is defined as $|\alpha| = \sqrt{\alpha^*\alpha}$).

Whenever a qubit is in a state described by amplitudes α_0 and α_1 that are both non-zero, we say that the qubit is in a *superposition* over the values "0" and "1". If we then *observe* the qubit, the probability that the observed outcome X is either 0 or 1 is then given by $\Pr[X = b] = |\alpha_b|^2 = |\langle\psi|b\rangle|^2$, and after observation, the superposition will have "collapsed" to the basis state corresponding to the observed value, $|b\rangle$.[3]

Likewise, a general, n-qubit quantum state may be written in terms of the computational basis states:

$$|\psi\rangle = \sum_{\forall i \in [2^n]} \alpha_i |i\rangle = \sum_{i=0}^{N-1} \alpha_i |i\rangle = \alpha_0 |0\ldots00\rangle + \alpha_1 |0\ldots01\rangle + \ldots + \alpha_{N-1} |1\ldots11\rangle \ ,$$

where the *amplitudes* α_i again satisfy $\sum_{\forall i\in[2^n]} |\alpha_i|^2 = 1$.

When expanded over the basis states, the inner product of two quantum states $|\psi\rangle = \sum_{\forall i\in[N]} \alpha_i |i\rangle$ and $|\varphi\rangle = \sum_{\forall j\in[N]} \beta_j |j\rangle$ may be written

$$\langle\psi|\varphi\rangle = \sum_{\forall i,j\in[N]} \alpha_i^* \beta_j \langle i|j\rangle = \sum_{\forall i,j\in[N]} \alpha_i^* \beta_j \delta_{ij} = \sum_{\forall i\in[N]} \alpha_i^* \beta_i \ .$$

We see that we have recovered the inner product of complex-valued vectors, which is just the usual inner product of real-valued vectors with an extra complex conjugation.[4] Thus we may apply the usual operational interpretation of the inner product, in which vectors are "perpendicular" when their inner product is 0, and "parallel" when it is 1. Together with the requirement that quantum states have unit norm, this means that an inner product of 1 implies that the two quantum states are actually identical, (up to the aforementioned unobservable phase factor).

When expanding quantum states in terms of basis states in the manner above, we will sometimes implicitly exclude terms for which $\alpha_i = 0$, writing instead

$$|\psi\rangle = \sum_i \alpha_i |x_i\rangle \coloneqq \sum_{\forall i\in\mathcal{I}} \alpha_i |x_i\rangle \ ,$$

where $\mathcal{I} = [\ell]$ indexes the set $\{|x\rangle \mid x \in [N] \ \wedge \ \langle\psi|x\rangle \neq 0\}$ for some $\ell \in \mathbb{N}$.

[3] The discernible effect of observations on quantum states has led to a century of heated philosophical debate over how quantum theory should be interpreted: The term "collapse" used here is typically associated with the conservative "Copenhagen interpretation", championed by Niels Bohr, and commonly characterized as taking a "shut-up-and-calculate" attitude towards the problem. Another popular interpretation is the many-worlds interpretation due to Hugh Everett, and favoured by father of quantum computing theory David Deutsch, among others. Instead of the quantum state "collapsing" to become "classical", this interpretation would say that both terms are ill-defined, and that the universe should instead be understood to be nothing but one big superposition state over all worlds that might have arisen from the initial conditions of the Big Bang; and that what actually happened when you made your observation, was that the quantum state describing the subsystem you were studying and the quantum state describing the subsystem containing *you* became correlated via the process known as *entanglement*. (In case you're wondering, I place my own bet on the *relational* interpretation due to Carlo Rovelli, which may be described as "many-worlds without the multiverse".)

[4] This is because inner products on any vector space must satisfy conjugate symmetry, $\langle x, y\rangle = \langle y, x\rangle^*$; see `https://en.wikipedia.org/wiki/Inner_product_space`.

Unitary operations. Since we are working over finite-dimensional Hilbert spaces, quantum operations, i.e. evolutions of quantum states, may be represented as $N \times N$-dimensional matrices. What properties should they have?

Clearly, any operation that we perform on a quantum state (*except* observing them!) should yield another valid quantum state as a result. Since quantum states must be unit-norm, this means that any operation we perform on them must preserve the norm. This fact turns out to be sufficient[5] to restrict the space of allowed operations to the unitary matrices, i.e. invertible square matrices U satisfying $U^{-1} = U^\dagger$.

Exercise 1. Prove that requiring operations to be norm-preserving is equivalent to restricting the set of legal operations to the set of unitary matrices.

We will use capital roman letters to refer to quantum operations, in reference to their matrix representation. Whenever the qubit(s) that the operation is being applied to is not clear from context, we will provide the index of the qubit(s) in the subscript, e.g. U_i for single-qubit operations, and $U_{i,j}$ for two-qubit operations, where $i, j \in [N]$ and $i \neq j$ (or more generally, the *registers* on which the operations are applied).

It is often useful to refer to operations in terms of the effect they have on the starting state, and so we define $|\psi\rangle \xrightarrow{U} |\varphi\rangle$ to mean that we perform the operation $|\psi\rangle \longrightarrow U|\psi\rangle = |\varphi\rangle$.

Some elementary unitary transformations, known as quantum *gates*, include

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad CX = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

Exercise 2. Write down "truth tables" for each of the above gates by writing down their effect on the (single-qubit resp. two-qubit) computational basis states. (E.g., $|0\rangle \xrightarrow{X} |1\rangle$.)

The "C" in CX is short for "controlled", where the value of the first qubit *controls* the operation on the second qubit. Any single-qubit operation U defines a two-qubit controlled version of the same operation, which we will refer to as CU.

Exercise 3. Starting with the two-qubit state $|\psi\rangle = |00\rangle$, apply the operations H_0 and $CX_{0,1}$. What are the possible outcomes of observing the resulting state, and what are their probabilities? Compare with the result of applying the single-qubit gates H_0 and H_1 instead.

Exercise 4. Write a new truth table for X, Z, and H, only now in terms of the alternative basis states $|+\rangle := \frac{|0\rangle+|1\rangle}{\sqrt{2}}$, $|-\rangle := \frac{|0\rangle-|1\rangle}{\sqrt{2}}$ instead. Compare with the truth table from Exercise 2. What has changed?

Exercise 5. Give the matrix representation of a unitary operation that acts on the alternative basis from Exercise 4 in the same way that CX acts on the computational basis.

References

NC10. Michael Nielsen and Isaac Chuang. *Quantum Computation and Quantum Information.* Cambridge University Press, second edition, 2010.

A Select Exercise Solutions

Exercise 1. Let U be such that, for all states $|\psi\rangle$, $\|U|\psi\rangle\| = \||\psi\rangle\|$. Using the definition of the norm, this gives us that $\left|(U|\psi\rangle)^\dagger U|\psi\rangle\right| = \left|\langle\psi| U^\dagger U |\psi\rangle\right| = |\langle\psi|\psi\rangle|$. Since this should hold for any $|\psi\rangle$, we must have that $U^\dagger U = \mathbb{1} = UU^\dagger$; the latter equality follows from the fact that, for all $|\varphi\rangle$ and $|\xi\rangle$, $|\langle\varphi|\xi\rangle| = |\langle\xi|\varphi\rangle|$. Thus, U satisfies the definition of an invertible matrix, with the inverse $U^\dagger$. Further, it is easy to see that the opposite also holds: any unitary matrix must be norm-preserving. Therefore, the set of norm-preserving operations on N-dimensional Hilbert space equal to the set of $N \times N$ unitary matrices.

[5] Here is a short proof of this fact: